RESEARCH
PAPER

DeepSeek-V2

A Strong, Economical, and Efficient Mixture-of-Experts Language Model

236B Total Parameters | 21B Activated | 128K Context

Multi-Head Latent Attention + DeepSeekMoE

2024

arXiv:2405.04434

The LLM Scaling Dilemma: Performance vs. Cost and Speed

Larger models improve quality but face prohibitive training costs and inference bottlenecks — a new architecture is needed.

The Scaling Problem

- **Training Cost Explosion**
  Dense models scale compute linearly with parameters.
  67B dense model already near practical cost limits.
- **KV Cache Bottleneck**
  Standard MHA stores full K and V per head per token.
  Cache grows linearly with sequence length and heads.
- **Inference Speed Degradation**
  Memory-bound decoding limits generation throughput.
  Larger models = slower token generation = higher cost.

Existing solutions sacrifice quality:
GQA/MQA reduce cache but degrade model performance.

DeepSeek-V2's Answer

- **Multi-Head Latent Attention (MLA)**
  Low-rank KV joint compression into compact latent vectors — 93.3% KV cache reduction.
- **DeepSeekMoE Architecture**
  Sparse activation: only 21B of 236B params per token — 42.5% training cost reduction.
- **Massive Throughput Gain**
  5.76x higher maximum generation throughput vs. DeepSeek 67B dense predecessor.

Result: Pareto-optimal performance-cost tradeoff
Better quality than MHA, cheaper than dense, faster inference.

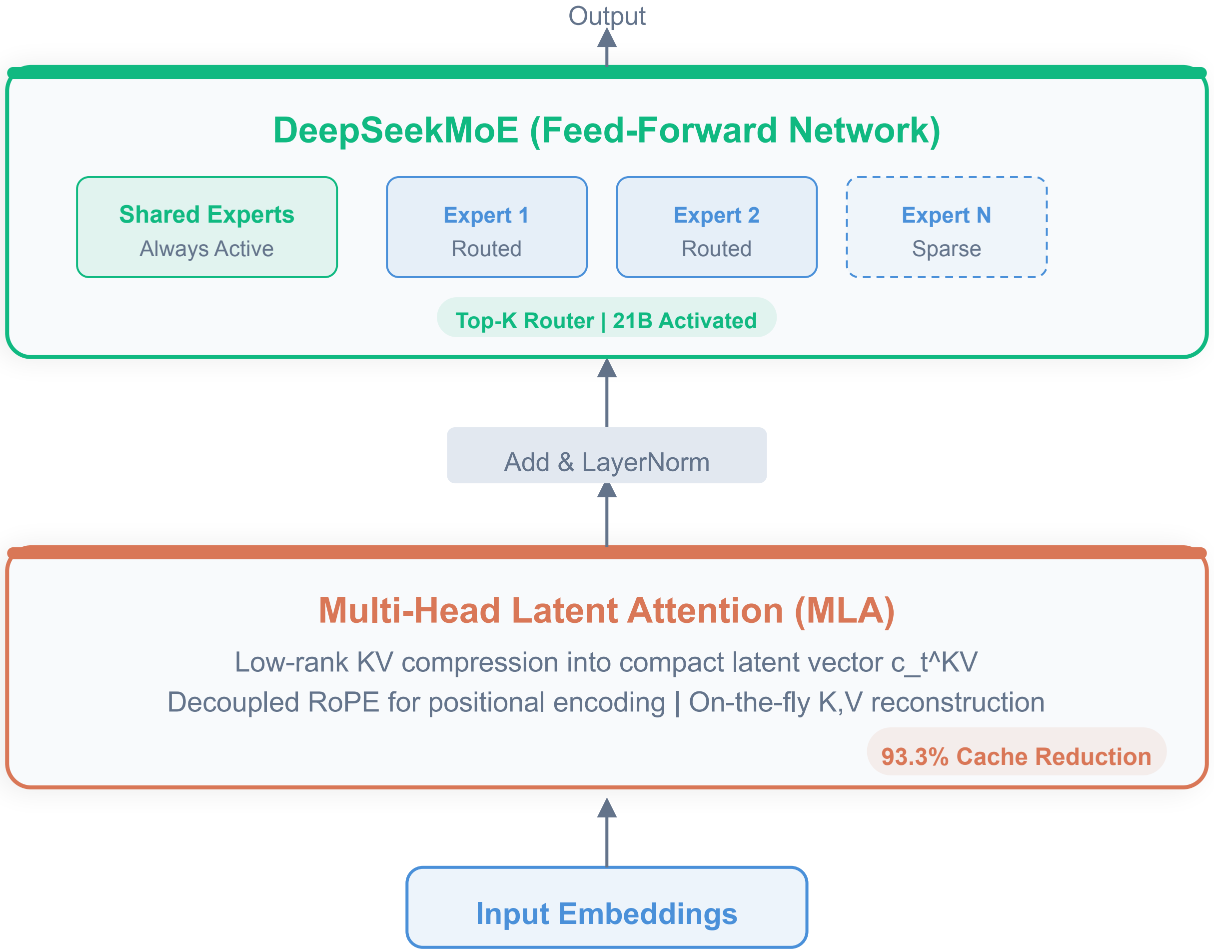
ARCHITECTURE

DeepSeek-V2: MLA + DeepSeekMoE in a Transformer Block

Two innovations replace standard attention and dense FFN, enabling 236B params with only 21B activated per token.

236B
Total Parameters

21B
Activated per Token



8.1T
Training Tokens

128K
Context Length

Source: DeepSeek-V2, arXiv:2405.04434

CORE INNOVATION 1

Multi-Head Latent Attention Compresses KV Cache by 93.3%

MLA achieves better performance than MHA while requiring dramatically less KV cache — a strictly superior alternative.

How MLA Works

1. **Compress K,V into Latent Vector**
   Joint low-rank projection: K,V --> compact c_t^KV
   Only c_t^KV is cached during inference (not full K,V)

2. **On-the-fly Reconstruction**
   Full K and V reconstructed via learned projection matrices during attention computation.

3. **Decoupled RoPE Encoding**
   Separate positional key for RoPE, decoupled from content-based key-value compression.

Result: Better performance than MHA + 93.3% less cache

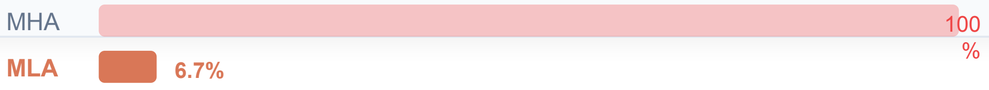
93.3% KV cache reduction vs. standard MHA

Attention Mechanism Comparison

Method	KV Cache	Performance	Tradeoff
MHA	Full (100%)	Best	Slow inference
GQA	Reduced	Degraded	Shared groups
MQA	Minimal	Worst	Single K,V head
MLA	6.7% of MHA	Best	No tradeoff

Key Insight:
MLA is the only method that improves both cache efficiency AND performance — making GQA/MQA strictly dominated solutions.

KV Cache Size (relative)



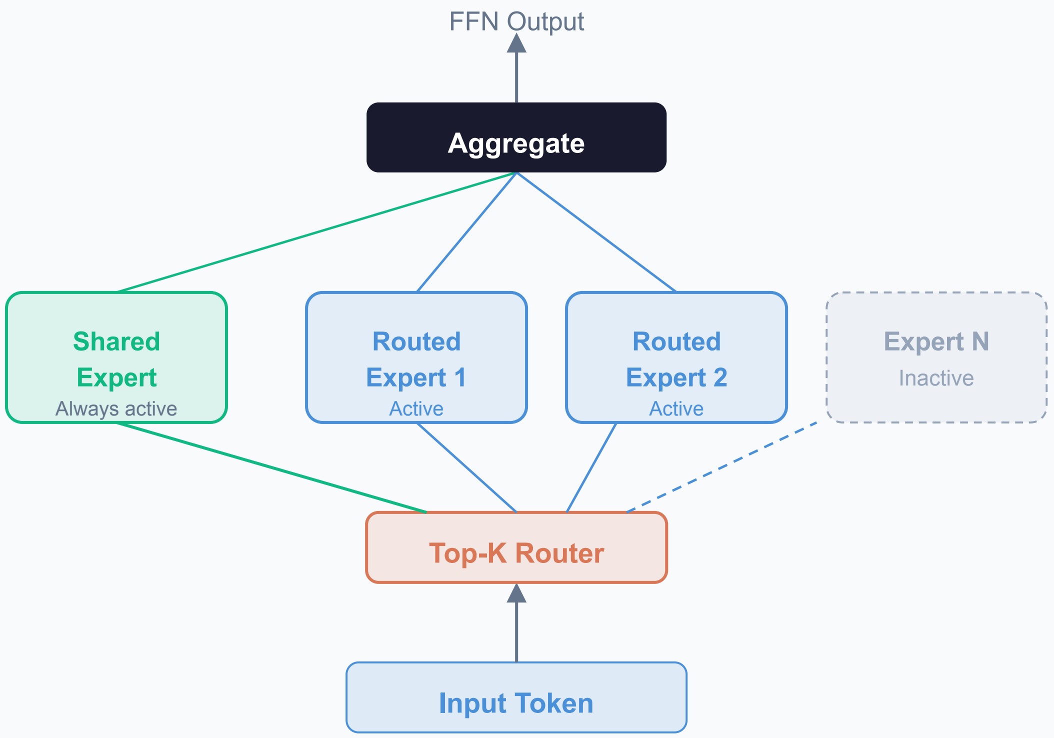
Source: DeepSeek-V2, arXiv:2405.04434

CORE INNOVATION 2

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Sparse activation of fine-grained experts enables 236B total params at 42.5% lower training cost than a 67B dense model.

MoE Architecture



Key Design Features

Fine-Grained Expert Segmentation

Experts split into smaller, more specialized units.
Increases specialization potential and routing precision.

Shared Expert Isolation

Dedicated shared experts (N_s) always activated to capture common knowledge across all inputs.

Device-Limited Routing

Each token routed to experts on at most M devices.
Reduces cross-node communication overhead.

Load Balancing & Token Dropping

Auxiliary losses enforce expert and device load balance.
Token-dropping in training prevents overflow (no impact on inference).

Result: 236B params trained at 42.5% lower cost than 67B dense

Source: DeepSeek-V2, arXiv:2405.04434

EFFICIENCY

42.5% Training Cost Reduction and 5.76x Throughput vs. DeepSeek 67B

DeepSeek-V2 achieves dramatic efficiency gains across training, memory, and inference vs. its dense predecessor.

DeepSeek 67B (Dense)	vs.	DeepSeek-V2 (236B MoE)

Training Cost Reduction

42.5%
lower training cost

DeepSeek 67B	100 %
DeepSeek-V2	57.5%

42.5% savings

KV Cache Reduction

93.3%
smaller KV cache

DeepSeek 67B	100 %
DeepSeek-V2	6.7%

93.3% savings

Generation Throughput

5.76x
higher max throughput

DeepSeek 67B	1x
DeepSeek-V2	5.76x

5.76x improvement

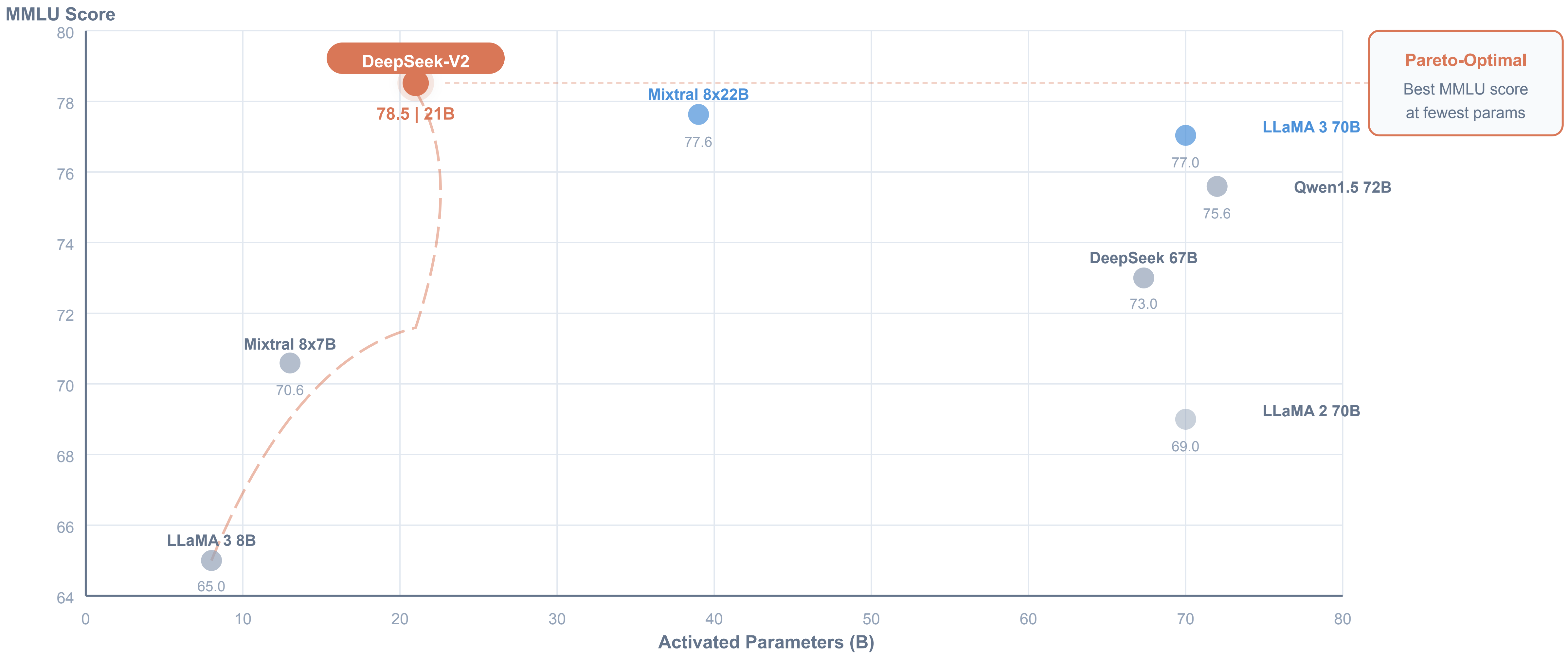
All metrics compared to DeepSeek 67B dense predecessor | 236B total params, 21B activated per token
Combined MLA + DeepSeekMoE innovations enable simultaneous improvements across all three dimensions.

Source: DeepSeek-V2, arXiv:2405.04434

BENCHMAR
K

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters

DeepSeek-V2 achieves the highest MMLU score (78.5) among open-source models while activating only 21B of 236B parameters.



Source: DeepSeek-V2, arXiv:2405.04434

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

After SFT + GRPO-based RL, DeepSeek-V2 Chat outperforms all open-source and most closed-source models in Chinese.

Alignment Pipeline

Pretraining (8.1T tokens) ◀ SFT (1.5M conversations) ◀ GRPO Reinforcement Learning ◀ DeepSeek-V2 Chat (RL)

ENGLISH EVALUATION

AlpacaEval 2.0

Length-controlled win rate

38.9%

Win rate vs. reference model

Competitive with top closed-source models in English open-ended evaluation.

MULTI-TURN EVALUATION

MT-Bench

Overall score (max 10)

8.97

8.97 / 10

Strong multi-turn conversational ability across diverse topics.

CHINESE EVALUATION

AlignBench

Overall score (Chinese)

7.91

7.91 / 10

Best among all open-source models
Outperforms most closed-source models

Source: DeepSeek-V2, arXiv:2405.04434

KEY TAKEAWAYS

Architectural Innovation Unlocks Efficient Scaling

1. **MLA: A Strictly Superior KV-Cache Solution**
   Better performance than MHA while reducing KV cache by 93.3% — makes GQA/MQA obsolete for inference efficiency.

2. **DeepSeekMoE: Economical Training at Scale**
   Fine-grained experts + shared expert isolation train a 236B-parameter model at 42.5% lower cost than a 67B dense model.

3. **Pareto-Optimal MMLU Performance**
   Only 21B activated parameters needed to match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.

4. **5.76x Throughput for Practical Deployment**
   Massive generation throughput improvement over DeepSeek 67B makes large-scale deployment economically viable.

5. **Strongest Open-Source Chinese Chat Model**
   DeepSeek-V2 Chat (RL) outperforms all open-source and most closed-source models on AlignBench (Chinese).